

Appendix: Generalization Experiments on Additional Datasets

1 Supplementary Results on UNSW-NB15

The UNSW-NB15 dataset was created to address the limitations of outdated benchmark datasets such as KDD98, KDDCUP99, and NSL-KDD, which do not adequately reflect modern network traffic and low-footprint attacks. The dataset contains a hybrid of real modern normal traffic and contemporary synthesized attack activities, comprising nine attack categories: Analysis, Backdoor, DoS, Exploits, Fuzzers, Generic, Reconnaissance, Shellcode, and Worms. To assess the generalizability of the proposed framework beyond the X-IIoTID dataset, we evaluate both the decoder and encoder on UNSW-NB15 using the same experimental protocol.

1.1 Decoder Performance

Table 1 presents the attack→normal fooling rate and MMD of the proposed decoder compared with gradient-based baseline attacks on the UNSW-NB15 dataset.

Table 1: Comparison of Fooling Rate and Distribution Quality (MMD) on UNSW-NB15.

Method	Fooling Rate ($\uparrow$)	MMD ($\downarrow$)
Decoder (Ours)	0.9216	0.0124
PGD-T	0.8804	0.5121
FGSM-T	0.7426	0.5108
JSMA-T	0.3303	0.4789
DeepFool-U	0.1160	0.5188
PGD-U	0.0299	0.5107
FGSM-U	0.0240	0.5089

The proposed decoder achieves the highest fooling rate of 0.9216 with the lowest MMD of 0.0124, consistent with the trends observed on X-IIoTID. Targeted gradient-based attacks such as PGD-T and FGSM-T also achieve relatively high fooling rates, but their MMD values remain substantially higher, indicating that the generated samples are distributionally farther from normal traffic. These results confirm that the proposed framework generalizes its ability to produce both effective and distribution-aligned adversarial samples across datasets.

1.2 Encoder Performance

Figure 1 compares the clean and mixed-set accuracy and macro-F1 scores of the proposed adversarially trained encoder against baseline classifiers on the UNSW-NB15 dataset.

On the UNSW-NB15 dataset, the proposed adversarially trained encoder achieves a mixed-set accuracy of 0.8613 and a mixed macro-F1 of 0.3951, outperforming the baseline encoder (No Adv.) which drops to a mixed accuracy of 0.8125 and macro-F1 of 0.4061 under adversarial conditions. While tree-based models such as Decision Tree and Random Forest achieve higher clean-set performance, their mixed-set macro-F1 scores degrade noticeably when adversarial samples are introduced. Notably, the proposed encoder maintains or improves its performance from clean to mixed settings, confirming that the adversarial training procedure provides robustness against decoder-generated evasion samples on this dataset as well.

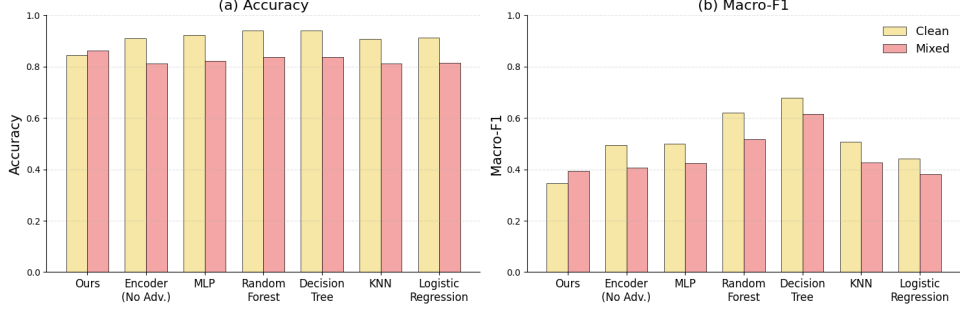


Figure 1: Encoder performance comparison on UNSW-NB15 under clean and mixed clean/adversarial evaluation sets. (a) Accuracy comparison. (b) Macro-F1 comparison.

2 Supplementary Results on TON_IoT

The TON_IoT dataset is a new generation of Industry 4.0/IoT and IIoT dataset designed to evaluate cybersecurity applications based on AI. It includes heterogeneous data sources collected from telemetry data of IoT and IIoT sensors, operating system logs, and network traffic, gathered from a realistic large-scale testbed at the Cyber Range and IoT Labs, UNSW Canberra. The dataset covers various attack types including DoS, DDoS, and ransomware across IoT/IIoT network environments, making it well-suited for evaluating the generalizability of intrusion detection frameworks under diverse and modern threat scenarios.

2.1 Decoder Performance

Table 2 presents the attack→normal fooling rate and MMD of the proposed decoder compared with gradient-based baseline attacks on the TON_IoT dataset.

Table 2: Comparison of Fooling Rate and Distribution Quality (MMD) on TON_IoT.

Method	Fooling Rate ($\uparrow$)	MMD ($\downarrow$)
Decoder (Ours)	0.9226	0.0573
PGD-T	0.6317	0.1888
JSMA-T	0.3706	0.1881
FGSM-T	0.0506	0.1922
PGD-U	0.0273	0.1823
FGSM-U	0.0267	0.1850
DeepFool-U	0.0165	0.1887

The proposed decoder achieves the highest fooling rate of 0.9226 with the lowest MMD of 0.0573 on the TON_IoT dataset, consistent with results observed on X-IIoTID and UNSW-NB15. While PGD-T achieve moderate fooling rates of 0.6773 and 0.6317 respectively, their MMD values remain substantially higher, indicating that gradient-based attacks fail to generate samples that are distributionally close to normal traffic. These results confirm that the proposed framework maintains its advantage in both attack effectiveness and distributional realism across heterogeneous IoT dataset environments.

2.2 Encoder Performance

Figure 2 compares the clean and mixed-set accuracy and macro-F1 scores of the proposed adversarially trained encoder against baseline classifiers on the TON_IoT dataset.

On the TON_IoT dataset, the proposed adversarially trained encoder achieves a mixed-set accuracy of 0.7518 and a mixed macro-F1 of 0.7142, outperforming all baseline models under the mixed evaluation setting. The baseline encoder (No Adv.) drops to a mixed accuracy of 0.5947 and macro-F1 of 0.6583 when adversarial samples are introduced, representing a substantial degradation. Tree-based models such as Random Forest and Decision Tree achieve high clean-set performance but show notable drops in mixed-set accuracy. Notably, the proposed encoder is the only model that maintains competitive

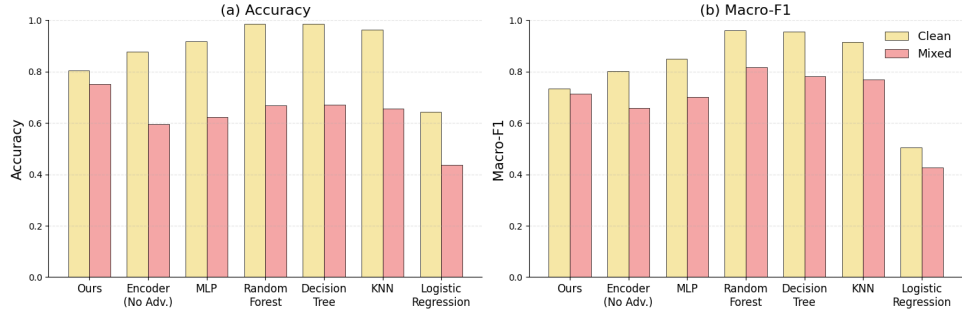


Figure 2: Encoder performance comparison on TON_IoT under clean and mixed clean/adversarial evaluation sets. (a) Accuracy comparison. (b) Macro-F1 comparison.

performance across both clean and mixed settings, confirming that the adversarial training procedure provides consistent robustness against decoder-generated evasion samples on this dataset as well.